



MONASH University

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DATA AND EVIDENCE IN ECONOMICS

**THE EFFECT OF INTERNATIONAL STUDENTS ON RENTAL PRICES IN NEW
SOUTH WALES AND VICTORIA BETWEEN 2019 - 2024**

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Introduction

In recent years, Australia has experienced a substantial increase in rental prices, particularly following the COVID-19 pandemic. At the same time, education remains one of the largest exports in Australia and continues to play an important role in economic activity. However, as the number of students recovered after the reopening of international borders, public and political debate are increasingly blaming the international students for worsening housing affordability and rising rents. Therefore, it is important to study whether changes in international students enrolments are significantly associated with changes in rental price to evaluate the claims surrounding migration, housing demand, and rental market conditions in the country.

This report investigates the effect of international student enrolments and rental prices in New South Wales (NSW) and Victoria (VIC) between 2019 and 2024. The research uses quarterly panel data from credible sources including the Australian Bureau of Statistics (ABS), the Reserve Bank of Australia (RBA) and the Australian Government Department of Education. Both rental prices and international student enrolments are measured using state-level data using median weekly rent and monthly enrollment respectively. The primary regression analysis implements a log-transformed first difference, a COVID-19 period dummy, state fixed effects, and controls for population growth and housing supply to examine the relationship between changes in international student enrolments and rental prices.

The results show little to no evidence of a reliable strong or statistically significant relationship between changes in international student enrolments and rental price, holding changes in population, changes in housing supply, the COVID-19 period effect, and time invariant differences between states constant. The main specification shows that the estimated relationship between international student enrolments and rental prices is extremely small and statistically insignificant. In addition, a set of robustness checks also provides similar conclusions across alternative specifications, which suggests that the main result is reasonably stable. The findings suggest that changes in international student enrolments do not appear to be a strong factor for short-run rental price changes at the state level.

Data

The analysis uses quarterly data for New South Wales (NSW) and Victoria (VIC) from 2019 Q1 to 2024 Q4, which results in 48 observations. Rental price, population, housing supply, and unemployment rates data were obtained from the Australian Bureau of Statistics (ABS), while the international student enrolment data were obtained from the Australian Government Department of Education. In addition to that, gross domestic product was obtained from the Reserve Bank of Australia (RBA).

Rental prices and unemployment rates were originally reported monthly but converted into quarterly observations. Both data were aggregated by averaging monthly values within each quarter. International student enrolments were originally reported as monthly year-to-date (YTD) but converted into quarterly observations. Population, housing supply and gross domestic product data were already reported quarterly. The regression analysis uses median weekly rent as the primary dependent variable while international student enrolments as the main explanatory variable. Population and housing supply are controls to account for broader demographic and housing market conditions. Although unemployment and GDP were initially considered, they were excluded from the main specification as these national-level macroeconomic variables may not align closely with state-level rental market data.

Table 1. Summary Statistics

<i>Variable</i>	<i>Mean</i>	<i>Std. Dev.</i>	<i>Min</i>	<i>Max</i>	<i>N</i>
GDP (A\$ million)	636,772.08	31,074.34	569,302	676,456	48
Unemployment Rate (%)	4.7	1.08	3.5	7	48
Housing Supply (Residential Stock)	3,083.38	283.21	2,662	3,484	48
International Student Enrolments	77,166.71	82,306.58	10,941	280,011	48
Population	7,457,177.25	794,035.65	6,518,716	8,536,892	48
Rental (\$ per week)	465.48	71.92	380	647	48

Note: Summary statistics for quarterly observations for New South Wales and Victoria from 2019Q1–2024Q4. Variables are reported in levels prior to log transformation and first differencing.

Table 1 presents summary statistics for the variables included in the dataset. Rental prices averaged \$465.48 per week across the sample period, while international student enrolments showed substantial variation, which indicates the disruptions caused by border closure and reopening. Population and housing supply shows comparatively lower variation over time which is consistent with their gradual structural growth.

Empirical Strategy

This analysis adopts a descriptive approach rather than a causal empirical approach. The objective is to examine whether changes in international student enrolments are significantly associated with changes in rental prices, rather than to estimate a causal effect. The estimated coefficient should be interpreted as conditional association after controlling for other variables.

The preferred specification is:

$$\Delta \ln(Rent_{it}) = \beta_0 + \beta_1 \Delta \ln(Students_{it}) + \beta_2 \Delta \ln(Population_{it}) + \beta_3 \Delta \ln(Supply_{it}) + \alpha_i + \lambda COVID_t + \varepsilon_{it}$$

where i indexes states, and t indexes quarters.

The dependent variable is the change in the log of median weekly rent, while the main explanatory variable is the change in the log of international student enrolments. Population and housing supply are included as controls to account for housing demand and market conditions. State fixed effects capture time-invariant differences between New South Wales and Victoria. The COVID-19 dummy variable accounts for pandemic-related disruptions.

The analysis uses log transformations as the variables are measured in different units that allows the coefficients to be interpreted approximately as percentage changes. The analysis also uses the first difference of the logged variables to focus on short-run changes and reduce the influence of the common upward trends in variables such as rents, population, and housing supply.

Despite holding constant population, housing supply, state fixed effects, and the COVID-19 period, the descriptive estimates may still be affected by other unobserved factors such as interest rates, inflation, housing policies, domestic migration, and other influences. In addition, the state level aggregation may also mask an important local variation in the housing market within two of the most expensive cities in Australia. Therefore, the results should be interpreted as descriptive evidence between international student enrolments and rental prices, rather than evidence of causality.

Results

Table 2 presents the preferred regression specification in respect to the relationship between changes in international student enrolments and changes in rental prices in New South Wales (NSW) and Victoria (VIC).

Table 2. Primary Regression Results

<i>Variable</i>	<i>Main Specification</i>
$\Delta \ln(\text{Students})$	0.0003 (0.001)
$\Delta \ln(\text{Population})$	1.3973 (0.948)
$\Delta \ln(\text{Supply})$	-4.1544 (2.349)
COVID	-0.0115 (0.005)
State Fixed Effects	Yes
Observations	46
R ²	0.565

Note: Entries represent coefficient estimates from regressions. The dependent variable is the quarterly change in the natural logarithm of median weekly rent. The specification includes controls for population growth and housing supply, state fixed effects, and a COVID-19 dummy variable. Standard errors are reported in parentheses.

The coefficient on the change in the log of international student enrolments is positive but extremely small. A 1% increase in international student enrolments is associated with approximately a 0.0003% increase in quarterly rental price growth, holding changes in population, changes in housing supply, the COVID-19 period effect, and time-invariant differences between states constant.

However, the estimate is statistically insignificant ($p = 0.812$) and economically negligible. There is little evidence of a meaningful relationship between short-run changes in international student numbers and rental prices during the sample period. The analysis suggests that broader demographic and housing market conditions may have played a more important role in rental price movements rather than changes in international student enrolments alone.

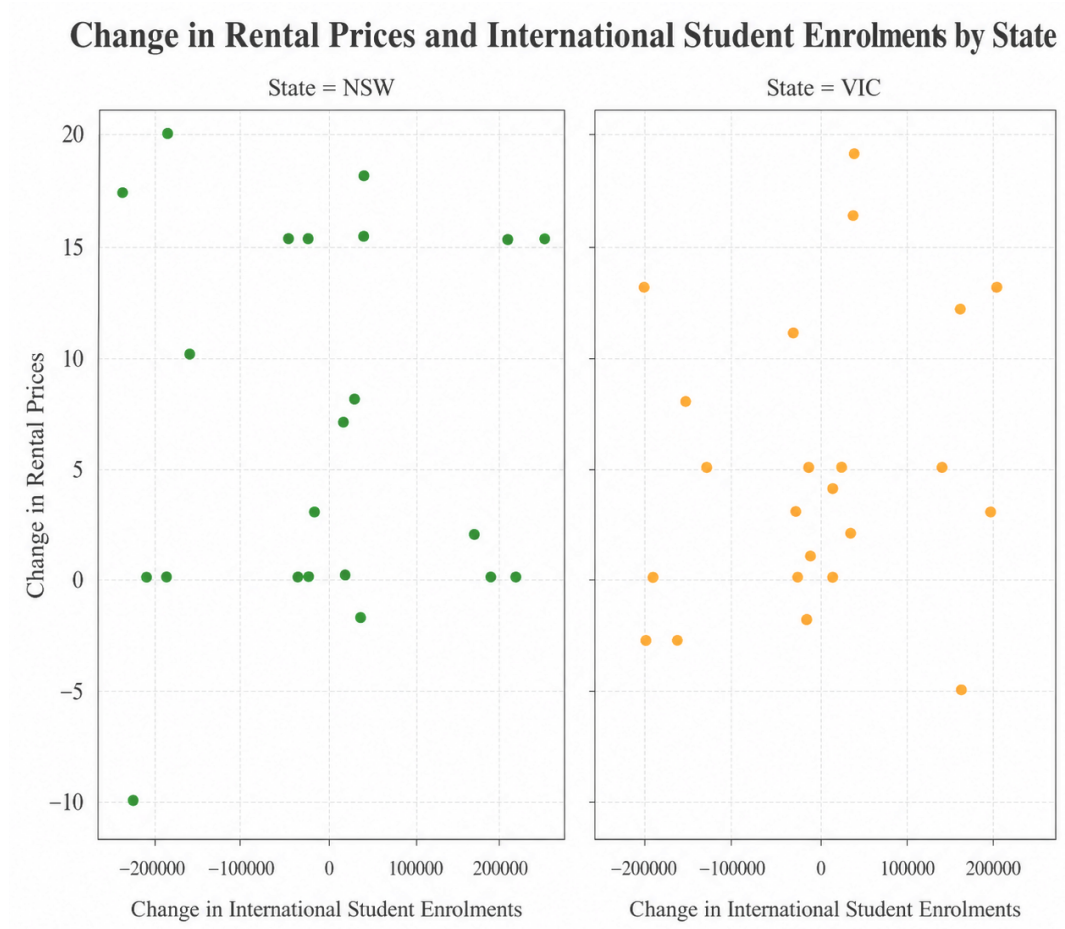


Figure 1: Scatterplot of Quarterly Changes in International Student Enrolments and Rental Prices

The figure above shows substantial dispersion in the relationship between quarterly changes in international student enrolments and rental price changes across both NSW and Victoria. The points do not form a clear upward or downward pattern, which suggests that the linear relationship between the two variables is weak or absent. In several quarters, large increases in student enrolments were associated with only small changes in rental prices, while relatively large rental price increases also occurred during periods where the number of international students fell. The pattern suggests that changes in international student numbers alone may not have been a major driver of short-run rental price movements during the sample period.

Robustness

The table presents several robustness checks to examine whether the estimated relationship between international student enrolments and rental price growth is sensitive to alternative specifications, samples, and modelling assumptions.

Table 3: Robustness Checks and Alternative Specifications

<i>Variables</i>	<i>Main</i>	<i>HC3</i>	<i>No COVID</i>	<i>NSW</i>	<i>VIC</i>	<i>Levels</i>	<i>Lagged</i>
$\Delta \ln(\text{Students})$	0.0003 (0.001)	0.0003 (0.001)	-0.0009 (0.001)	-0.0003 (0.002)	0.0009 (0.002)	-0.002 (0.004)	-0.0009 (0.001)
$\Delta \ln(\text{Population})$	1.3973 (1.009)	1.3973 (1.009)	2.987 (0.707)	1.4137 (2.323)	1.6066 (1.26)	4.9648 (0.435)	1.2495 (0.886)
$\Delta \ln(\text{Supply})$	-4.1544 (2.595)	-4.1544 (2.595)	-3.8533 (2.475)	-4.3678 (4.338)	-2.8857 (3.863)	-0.6652 (0.286)	-3.4096 (2.329)
COVID	-0.0115 (0.005)	-0.0115 (0.005)	— —	-0.012 (0.007)	-0.0099 (0.008)	-0.0067 (0.01)	-0.0135 (0.004)
State Fixed Effects	Yes	Yes	Yes	No	No	Yes	Yes
First-Differenced Logs	Yes	Yes	Yes	Yes	Yes	No	Yes
Robust Standard Errors	No	Yes	No	No	No	No	No
Sample Restriction	Full	Full	Full	NSW	VIC	Full	Full
Observations (N)	46	46	46	23	23	48	44
R ²	0.565	0.565	0.504	0.558	0.583	0.975	0.591

Note: The outcome variable is $\Delta \ln(\text{Rent})$ except in column (6), where the outcome is $\ln(\text{Rent})$ in levels. Column (6) uses logged-level variables instead of first-differenced logged variables. All specifications control for population growth and housing supply. Column (2) reports HC3 robust standard errors. Column (3) removes the COVID dummy. Columns (4) and (5) estimate the model separately for NSW and VIC. Column (6) re-estimates the model without first differencing. Column (7) includes a one-quarter lag of $\Delta \ln(\text{Students})$. State fixed effects are included where indicated. Standard errors are reported in parentheses.

The robustness checks suggest that the main finding is reasonably stable and robust across all specifications. The estimated effect of international student enrolments on rental prices remains economically small and statistically insignificant. Although the sign of the coefficient changes across some specifications, there is no consistent evidence of a strong positive or negative relationship once broader demographic and housing factors are controlled for.

The HC3 robust standard error specification in Column (2) produces more reliable standard errors in small samples which is identical to the main model in Column (1), which suggests

that heteroskedasticity is not materially affecting inference. The main finding therefore remains credible under alternative standard error assumptions.

Removing the COVID dummy in Column (3) changes the coefficient from slightly positive to slightly negative and lowers the R^2 from 0.565 to 0.504. This suggests that the COVID period captures an important common shock affecting rental markets. However, the student enrolment effect remains economically small, indicating that the baseline result is not entirely dependent on the specific pandemic adjustment.

The NSW-only and VIC-only regressions in Columns (4) and (5) also produce very small coefficients with different signs across states. While this indicates some sensitivity across subsamples, neither state shows a statistically meaningful relationship. This suggests that the main finding is not driven entirely by one state.

The log-level specification in Column (6) produces the largest change, with a much higher R^2 of 0.975 and a larger negative coefficient on student enrolments. This likely reflects strong common upward trends in rents and macroeconomic variables over time rather than a stronger economic relationship. Because the levels model does not remove persistent time trends, the result supports the use of first-differenced logs in the main specification to obtain a more conservative estimate of the short-run relationship.

Finally, the lagged specification in Column (7) still produces a very small and statistically insignificant coefficient, suggesting little evidence that rental prices respond to changes in student enrolments with a one-quarter delay.

Discussion, limitations, conclusions

The analysis finds little evidence of a strong and reliable relationship between changes in international student enrolments and rental prices in New South Wales and Victoria between 2019 and 2024. The estimated coefficient of international student enrolments is extremely small and statistically insignificant holding constant population growth, housing supply, state fixed effects, and the COVID-19 period. In addition, a set of robustness checks also provides similar conclusions, as the estimated coefficient remained small across alternative specifications despite changing sign in some models. The findings suggest that changes in international student enrolments do not appear to be a strong factor for short-run rental price changes at the state level. The analysis identifies conditional associations rather than a causal

effect and therefore should be interpreted carefully due to several limitations that should be acknowledged.

First, the analysis uses aggregate state-level data, which may not fully capture the true housing market where there is a high number of international students near universities and city centres. In addition to that, potential omitted variable bias from broader economic and housing factors may influence both international students and rental price, as the analysis uses a descriptive approach rather than a causal empirical approach. Furthermore, although the use of first-difference in the preferred specification helps reduce common upward trends, it may overlook long run relationships within the housing market.

In conclusion, the analysis finds limited evidence of a strong association between international student enrolments and rental prices in New South Wales and Victoria between 2019 and 2024. Future research using suburb-level rental data and university-area student enrolments data as well as stronger identification strategies may provide better insight into housing market pressures.

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